

8 × 40-channel Gang Connector Using Snap-Beam Connectors in Co-packaged Optics

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Abstract A low-loss 8 × 40-channel gang connector using Snap-Beam connectors has been developed, enabling high-density simultaneous connection in Co-packaged Optics. Low insertion loss ≤ 0.7 dB and return loss ≥ 36 dB were verified for fibre-to-fibre connections. ©2026 The Author(s)

Introduction

Co-packaged Optics (CPO) is a key enabler for higher capacity and lower power consumption in next-generation data centres [1]. Figure 1 schematically illustrates a switch based on CPO. In CPO, photonic integrated circuits (PICs) based on silicon photonics are placed close to a large-scale integration (LSI) such as an application-specific integrated circuit (ASIC) and perform transmission and reception of optical signals. For the input and output of light to and from a PIC, an optical coupling device for a PIC-to-fibre connection is necessary. A configuration with optical interposers between PICs and optical fibres, such as a planar lightwave circuit (PLC) or a glass substrate with waveguides [2], is one of the ways to realize waveguide pitch conversion and mode field conversion between a PIC and an optical fibre. Therefore, optical connectivity between a wide variety of waveguides and optical fibres is a key technology for realizing CPO. In the package process of a CPO substrate, to avoid the complexity of optical fibre handling, the detachability of optical fibre side connectors is required for PIC-to-fibre connections [3, 4]. In addition, since a CPO substrate is packaged by a reflow process, reflow compatibility of optical connectors is also required. In CPO, fibre-to-fibre connections are also required within equipment to connect the package substrate and the front panel, as shown in Figure 1. To achieve high-density optical fibre wiring and connection, optical connectors must be compact, easy to attach and detach, and resistant to dusty environments. To satisfy these requirements, we have been developing an ultra-compact detachable optical connector, the Snap-Beam connector (SB connector), which couples with a collimated beam and is automatically mated by magnetic force. We have already developed 12-channel, 250 μm -pitch SB connectors and verified low insertion loss < 0.4 dB and return loss > 38 dB with randomly mated connectors [5]. We also demonstrated temperature cycle tests in fibre-to-fibre connections [5] and compatibility with a 260 °C reflow process in a waveguide-to-fibre connection [6]. In addition, to realize higher-density optical connections, a 40-channel, 127 μm -pitch SB connector was

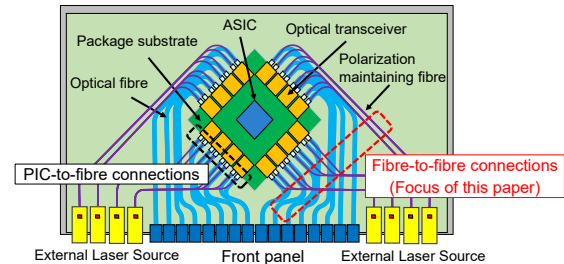


Fig. 1: Schematic illustration of switch based on CPO.

fabricated. For the 40-channel SB connector, low insertion loss < 0.4 dB and 0.6 dB was verified for fibre-to-fibre and waveguide-to-fibre connections, respectively [7].

In this paper, to realize high-density fibre-to-fibre connections with excellent operability in CPO, we fabricate an 8 × 40-channel gang connector using SB connectors and evaluate its insertion loss and return loss in fibre-to-fibre connections.

Design of SB Connector

Figure 2 shows the structure of the SB connector for a fibre-to-fibre connection. The fibre arrays (FAs) are connected with MT connector-like positioning parts. The positioning parts are made of a polyphenylene sulfide (PPS) based material and are manufactured by injection moulding with high productivity. The lens arrays (LAs) are precisely positioned with the positioning parts with reference to the guide mechanism and fixed, and the FAs are actively aligned and fixed to the rear surface of the LAs respectively. The light emitted from the optical fibre is collimated and focused by a pair of LAs and coupled to the core of the opposing optical fibre. By expanding the beam diameter by the lens, the tolerance for misalignment between the positioning parts is relaxed. In this study, the beam diameter of the collimated beam is designed to be 60 μm , the misalignment at which the coupling loss deteriorates by 0.2 dB is enlarged to about 6.5 μm from about 1 μm in a connection of standard single mode fibres.

In addition, by employing a non-contact connection instead of a physical contact (PC) connection conventionally applied to multi-fibre connectors, a strong spring mechanism for a PC connection and a robust connector housing can be

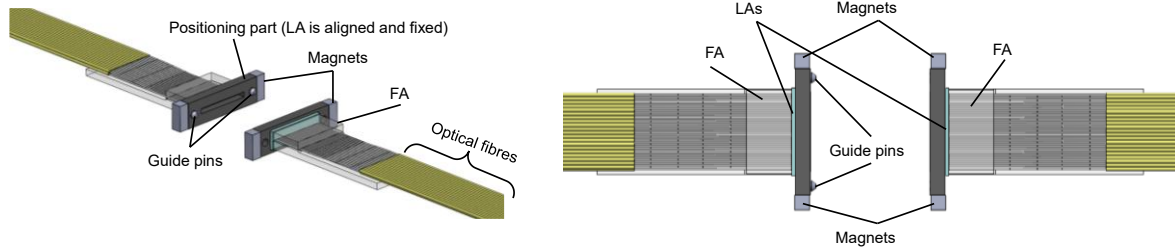
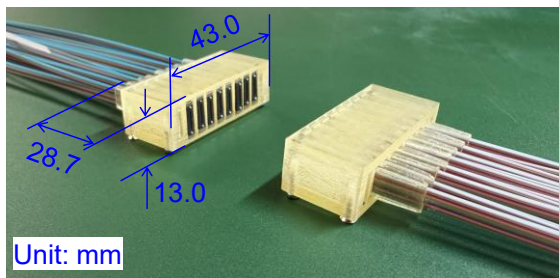


Fig. 2: Structure of SB connector for fibre-to-fibre connection.

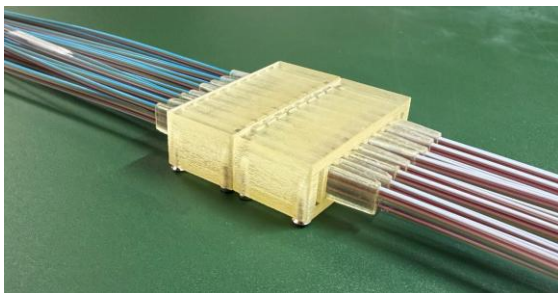
eliminated, and the connector terminal can be downsized. In the proposed structure, magnets are fixed to both sides of the positioning part, and a mating force is applied to the connector terminals by the magnets. When the opposing connector terminals come close to each other within a range of a few millimetres, the connector pairs can be connected automatically by the magnetic force. Hence, the workability of the edge connection in a limited space is secured. The magnetic force is designed to be approximately 2.2 N.

Design of 8 × 40-channel Gang Connector

Because the SB connector enables high-density connectivity without constraints on the mating force, it is well suited for application in a gang connector. Figures 3 and 4 show the appearance of an 8 × 40-channel gang connector and 40-channel SB connectors applied to the gang connector, respectively. The size of the gang connector terminals is 13.0 mm in height, 43.0 mm in width, and 28.7 mm in length. In the gang connector, 8 pairs of 40-channel SB connectors are mated simultaneously, enabling 320-channel high-density connectivity and improved operability within CPO.



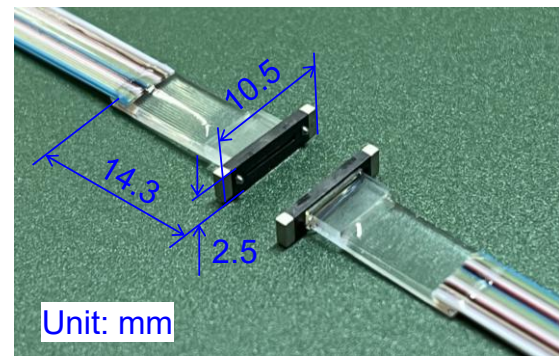
Before connection



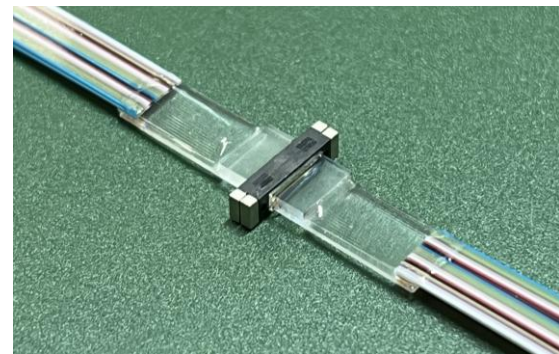
After connection

Fig. 3: 8 × 40-channel gang connector.

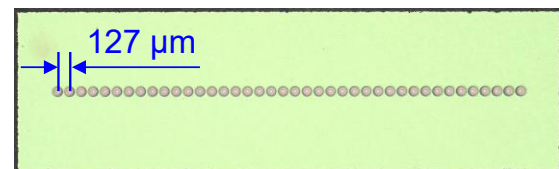
Figure 4 shows a 40-channel SB connector for an FA-to-FA connection and the LA used therein. The pitch of the optical fibres in the FA and the lenses in the LA is 127 μm . The optical fibres comply with ITU-T G.657, and their mode field diameter (MFD) is 8.6 μm at a wavelength of 1310 nm. The beam diameter expanded by the LAs is 60 μm .



Before connection



After connection



End face of 40-channel lens array

Fig. 4: 40-channel SB connector.

Optical Performance of the Gang Connector

Figures 5 and 6 show histograms of the insertion loss and the return loss of the gang connector for a single mating operation at a wavelength of 1310 nm, respectively. In a single mating operation, 8 pairs of 40-channel SB connectors are mated, and measurement values for a total of 320 channels were obtained. Low insertion

loss ≤ 0.7 dB and return loss ≥ 36 dB were demonstrated. In addition, Figure 7 shows the insertion loss and its variation over 5 mating cycles. Small loss variation $\leq \pm 0.06$ dB was confirmed. Consequently, the gang connector was demonstrated to be suitable for fibre-to-fibre connections in CPO, offering excellent operability, high-density, and low loss.

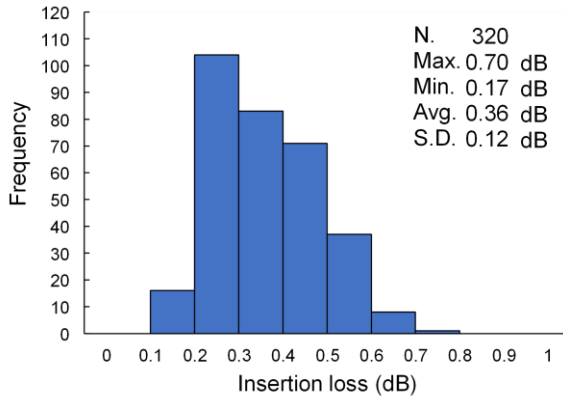


Fig. 5: Insertion loss of 8 × 40-channel gang connector.

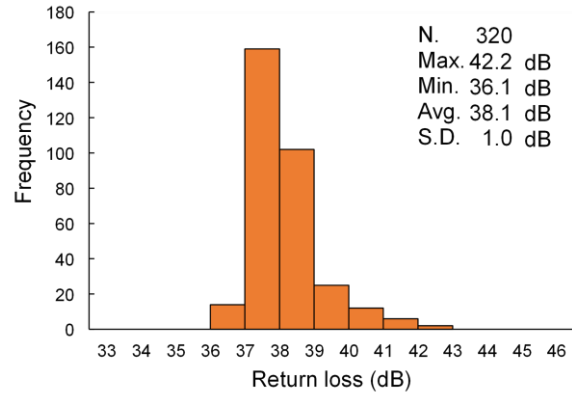


Fig. 6: Return loss of 8 × 40-channel gang connector.

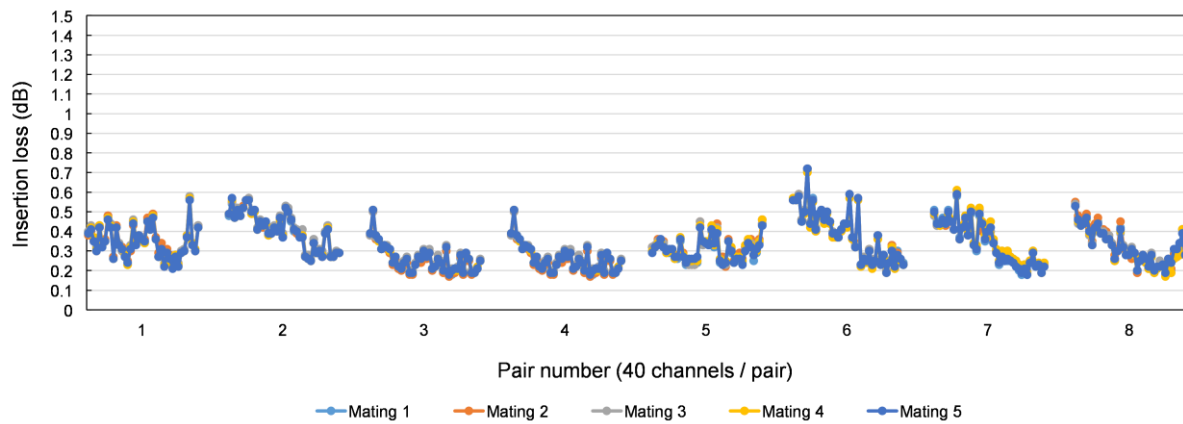


Fig. 7: Insertion loss and its variation of 8 × 40-channel gang connector over 5 mating cycles.

Conclusions

The SB connector developed by the authors employs expanded-beam coupling, enabling high-density connectivity without limitations imposed by the mating force, and is therefore well suited for application in gang connectors. An 8 × 40-channel gang connector using SB connectors was fabricated to realize high-density fibre-to-fibre connections with excellent operability in CPO. Low insertion loss ≤ 0.7 dB with a variation $\leq \pm 0.06$ dB and return loss ≥ 36 dB were demonstrated, confirming the effectiveness of the proposed gang connector for high-density fibre-to-fibre connectivity in CPO applications.

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